

Rail Engineering Mathematics

Tutorial Sheet 3

Questions

1. A cylinder is filled with water to a height of 50 cm. The cylinder has a hole at the bottom which is covered with a stopper. The stopper is released at time $t = 0$ seconds and is allowed to empty. The following data shows the height of the water in the cylinder at different times:

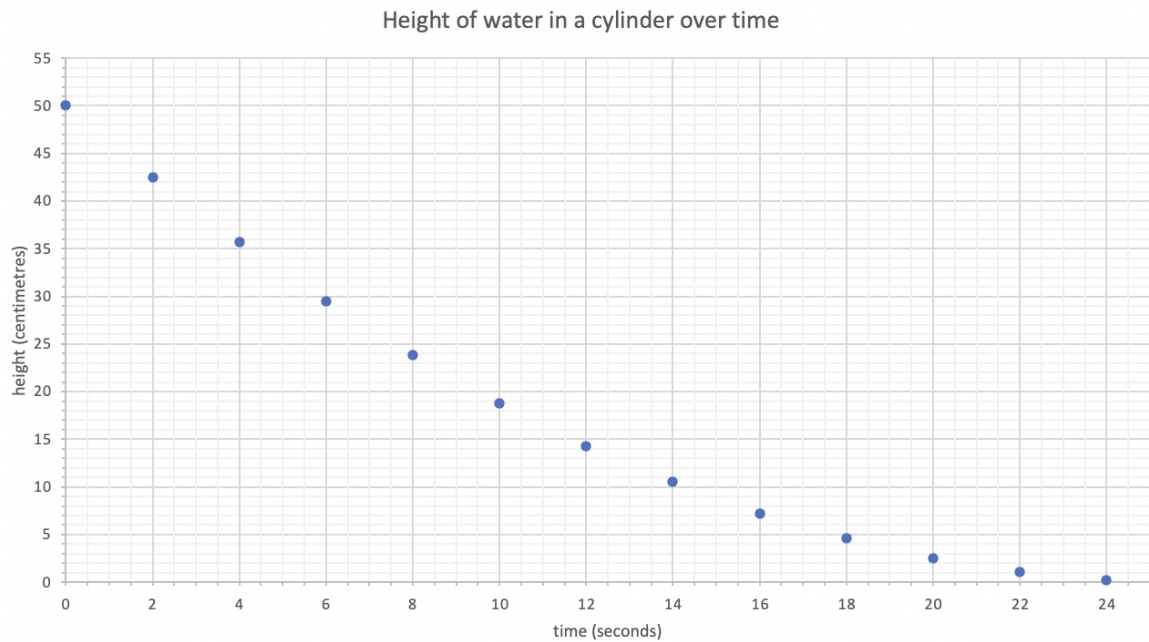
Time (seconds)	Height (cm)
0	50
2	42.5
4	35.7
6	29.5
8	23.8
10	18.8
12	14.3
14	10.5
16	7.2
18	4.6
20	2.5
22	1.1
24	0.2

Depicted in the graph below is the same data:

Assume that the relationship between the height of the water and the time is linear.

(a) With a ruler, draw a linear line of best fit on the graph and determine the equation of the line.

(b) Using the equation obtained in part (a) calculate the height of the water at $t = 1$ s, $t = 12$ s and $t = 24$ s.



(c) What can you conclude about the linear fit in comparison to the original data?

(d) Suggest a better model for the data, determine the equation by hand and test its viability plot plotting the equation and original data on the same axes (This data is available on Blackboard in the file “Tutorial3DataQ1.xlsx”).

- A pulley is driven by a flat belt with an angle of lap $\theta = \frac{2\pi}{3}$. T_2 is the tension, in N, at one end of the belt. The tension at the other end of the belt remains fixed. Data has been collected, via a series of experiments, by increasing the angle of lap and measuring the tension.

Using MATLAB’s `cftool` feature, determine a model that best describes the data (This data is available on Blackboard in the file “Tutorial3DataQ2.xlsx”).

Use MATLAB's `cftool` to fit curves to the data sets below, then state the relationship you were able to establish between the variables. You may need to refer to the “Interactive Function Explorer” on the module Blackboard site when trying to decide on a suitable function for each problem.

Note: some of the data permit numerous relationships to be explored and established. All original data can be found in individual spreadsheet files.

3. An experiment on the length L of a steel wire under various loads F yields the following data:

F (N)	L (m)
20	2.00042
40	2.00086
60	2.00132
80	2.00176
100	2.00219

4. Physical tables give the variation of atmospheric pressure p with changing altitude z as follows (p_0 represents sea-level pressure):

z (km)	p (% of p_0)
0	100
5	53
10	26
14	14
20	5.4
24	2.9
30	1.2

5. Johannes Kepler spent much time measuring the rotation period T of the planets about the Sun and their mean distances r from the Sun:

Planet	r (Gm)	T (days)
Mercury	57.9	88
Venus	108.2	225
Earth	149.6	365
Mars	227.9	687
Jupiter	778.3	4329
Saturn	1427	10753
Uranus	2870	30660
Neptune	4487	60150
Pluto	5907	90670

6. An experiment has been carried out to determine the relationship between the mean flow velocity V (m/s) and the pressure drop per unit length ΔP (Pa/m) across a short section of a smooth circular pipe. For laminar flow in a constant-diameter pipe the pressure drop is expected to be directly proportional to the mean velocity. The experiment validates a pipeline sensor design for low-speed flows.

A steady airflow is driven through the test pipe in a wind-tunnel rig. At each set mean velocity V , the pressure drop per unit length ΔP is measured using a calibrated differential pressure transducer. To account for small flow disturbances and sensor noise, four repeated measurements are taken at each velocity. All data are recorded in the spreadsheet “Tutorial3DataQ6.xlsx”.

(a) Plot all data points on a single scatter graph with the pressure drop per unit length ΔP on the vertical axis and the mean velocity V on the horizontal axis. Discuss the physical meaning of the trend shown in the graph.

(b) Using MATLAB’s Curve Fitting Tool (cftool), fit the following mathematical models to all data points:

- A Linear model
- A Quadratic model
- A Power model

Report the equation and the R^2 value for each of the three models.

(c) Plot all raw data points and the three fitted models from part (b) on a single graph, over the velocity range $V = 0$ to $V = 1.2$ m/s.

Based on both the statistical indicator (R^2) and the physical implications of each fitted curve up to 1.2 m/s, decide which mathematical model best represents the experimental data. Discuss the strengths and weaknesses of your chosen model and critically evaluate limitations of using it for extrapolation beyond the measured range.